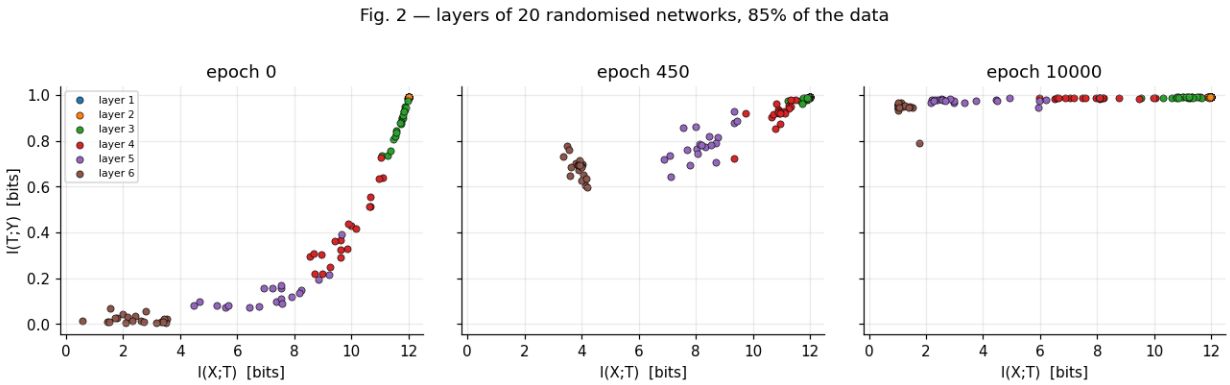
Reproducing Opening the Black Box of Deep Neural Networks via Information

Scientific reproduction and critical evaluation of information-plane dynamics



Information-plane snapshots from the submitted 20-network, 10,000-epoch notebook run.

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Abstract

This project reproduces and critically evaluates the information-plane experiments of Shwartz-Ziv and Tishby (2017) using a transparent NumPy implementation. The submitted long preset enumerates all 4,096 twelve-bit input patterns, trains 20 independently initialized 12-10-7-5-4-3-2 networks for 10,000 epochs with plain SGD, and estimates mutual information after quantizing layer activations into 30 bins. The reproduction recovers the central descriptive pattern: deeper layers first gain label information and later lose binned input information. The operational gradient signal-to-noise ratio crosses below one near epoch 84. Four of six measured layers satisfy the report's practical compression criterion. Converged deep layers have small vertical gaps to the Information Bottleneck frontier, but wide layers retain substantial horizontal excess information. Depth shortens threshold-crossing time up to three hidden layers and then becomes non-monotone. ReLU and bin-count controls show that the magnitude of measured compression depends on the estimator. These results support the information plane as a useful descriptive framework while limiting claims about continuous mutual information, causal diffusion, global IB optimality, and universal depth acceleration.

1 Introduction

Prediction accuracy reveals whether a network produces correct outputs, but it does not describe how hidden layers reorganize information during training. The assigned paper proposes a two-coordinate description for every layer: I(X;T), the information a representation retains about the input, and I(T;Y), the information it retains about the target. Plotting these quantities across epochs produces an information plane in which fitting appears as an upward movement and compression appears as a leftward movement.

The purpose of this reproduction is not to treat the published figures as illustrations to redraw. The notebook reconstructs the task, neural network, optimizer, gradient diagnostic, quantized mutual-information estimator, Information Bottleneck comparison, and controlled experiments. Every claim below is tied to a saved notebook output. Observed binned information is reported directly, while mechanistic and general claims remain qualified.

Original hypotheses and reproduction questions

Paper claim	Reproduction question	Assessment rule
Two SGD phases	Do layers first fit labels and later lose binned input information?	Inspect trajectories and layer-wise peak-to-final drops.
Noise-dominated late phase	Does the gradient SNR fall below one and remain low for most epochs?	Use the notebook's operational SNR definition.
IB convergence	Are endpoints close to the IB frontier in both coordinates?	Report vertical gap and horizontal excess separately.
Depth acceleration	Do additional hidden layers reduce the threshold-crossing time?	Compare epochs at a fixed fraction of available label information.

2 Information-theoretic foundations

Entropy, conditional entropy, and mutual information

For a discrete random variable, Shannon entropy measures uncertainty in bits.

$$H(X) = -\Sigma_x\, p(x)\, \log_2 p(x)$$

Conditional entropy measures uncertainty that remains after observing a representation.

$$H(Y|T) = \Sigma_t\, p(t)\, H(Y|T=t)$$

Mutual information is the reduction in uncertainty provided by the other variable.

$$I(X;T) = H(T) - H(T|X) \qquad I(T;Y) = H(Y) - H(Y|T)$$

The two coordinates have distinct meanings. I(X;T) measures distinguishable input information under the implemented quantizer. I(T;Y) measures population-level label relevance using the known stochastic label rule. Accuracy cannot replace either coordinate.

Data processing and changing parameters

At one fixed epoch, the network forms the Markov chain Y to X to T1 to T2 and onward to the prediction. The Data Processing Inequality limits the information a downstream layer can create about an upstream variable. This comparison applies across layers with fixed parameters. It does not require a layer's mutual information to decrease from one training epoch to the next, because SGD changes the mapping itself.

Information Bottleneck

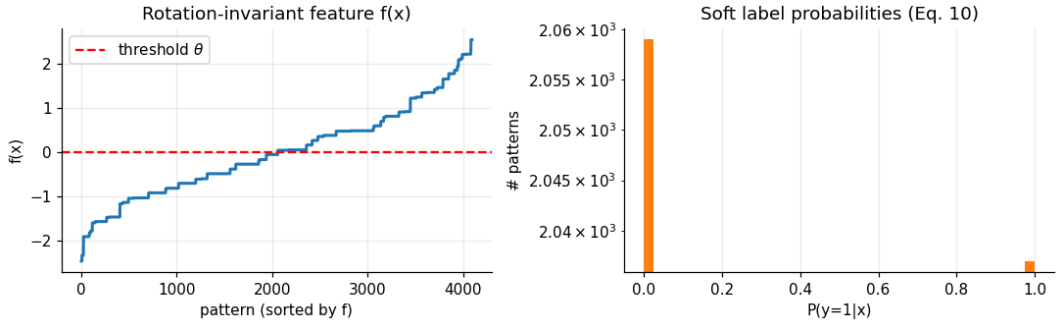
$$L_IB = I(X;T) - \beta\, I(T;Y)$$

The multiplier beta controls the trade-off between input compression and predictive relevance. The reproduced network is optimized with cross-entropy, not with this Lagrangian. The IB frontier is therefore a theoretical comparison curve. A learned layer can be vertically close to the curve while still retaining far more input information than necessary at the same label information.

Quantization and scope of the estimates

The hidden activations are continuous. The notebook maps each neuron value to an equal-width bin and concatenates the bin indices into a discrete layer code. For a fixed network and quantizer, this code is a deterministic function of X, so H(T|X)=0 and I(X;T)=H(T). A decrease in this discrete entropy indicates more quantization collisions. It must not be reported as an identical decrease in the mutual information of the underlying continuous vectors.

3 Dataset and implementation



The icosahedron-based feature, label calibration, and conditional label probabilities used by the submitted notebook.

Synthetic distribution

Each input X contains 12 binary indicators associated with the vertices of an icosahedron. Enumerating all assignments yields 2^12 = 4,096 equally likely patterns. A rotation-invariant spherical-harmonic feature partitions the patterns into 64 feature classes. A logistic rule with gain gamma=85.3 assigns p(Y=1|X). The threshold is calibrated so that P(Y=1)=0.497 and I(X;Y)=0.990 bits. Training labels are sampled from this conditional rule, whereas information is evaluated over the complete input distribution.

Neural network and SGD

Component	Submitted implementation
Architecture	12-10-7-5-4-3-2
Nonlinearity	Five tanh hidden layers; two-unit softmax output
Loss	Sampled-label cross-entropy
Optimization	Plain SGD, learning rate 0.05, batch size 1024
Ensemble	20 independently initialized networks trained in parallel
Schedule	10,000 epochs, seed 0, 119 saved frames
Information estimator	30 equal-width bins in the baseline
Evaluation	All 4,096 patterns at each information snapshot

The forward pass applies affine maps and tanh nonlinearities, followed by a softmax. Manual backpropagation propagates the cross-entropy gradient through every matrix. The SGD update is theta_(k+1) = theta_k - eta times the minibatch gradient estimate. Hidden activations and gradient statistics are extracted only at saved snapshots, which separates optimization from measurement and plotting.

Gradient signal-to-noise statistic

For each layer, the notebook compares the norm of the mean minibatch gradient with the norm of its fluctuations and aggregates layer ratios with a geometric mean. The first post-peak crossing below one defines the operational transition. This definition is reproducible but does not, by itself, prove Brownian diffusion.

4 Main information-plane reproduction

Fig. 2 — layers of 20 randomised networks, 85% of the data

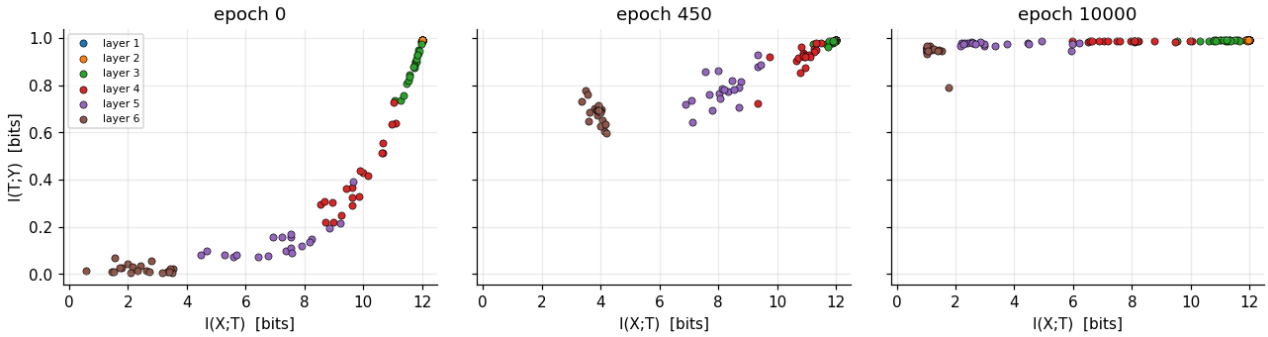


Figure 2 reproduction: layer positions for 20 networks at early, middle, and final training stages.

The ensemble begins with little label information in deep layers. During fitting, I(T;Y) increases. After the transition, narrower layers move toward lower I(X;T) while remaining high vertically. At epoch 10,000, the output mean is approximately I(X;T)=1.15 bits and I(T;Y)=0.943 bits. The widest two layers remain near the 12-bit input ceiling. The figure therefore supports a layer-dependent fitting-to-compression pattern under the 30-bin estimator.

Training-set-size trajectories

Fig. 3 — information-plane trajectories (green = drift→diffusion transition)

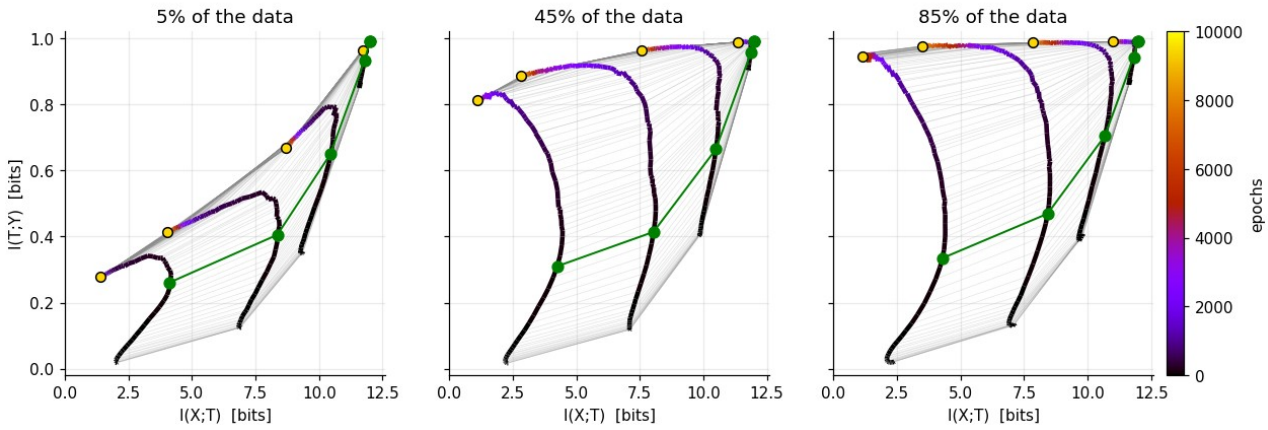


Figure 3 reproduction: information-plane trajectories across selected training fractions.

All trajectories use the same population evaluation. Their late endpoints differ because each optimizer sees a different fraction of the input universe. Larger training sets retain more target information at the output. The green markers locate the SNR-defined transition. The result links sample coverage with population-level relevance, but it does not show that compression causes generalization.

5 SGD dynamics and layer-wise compression

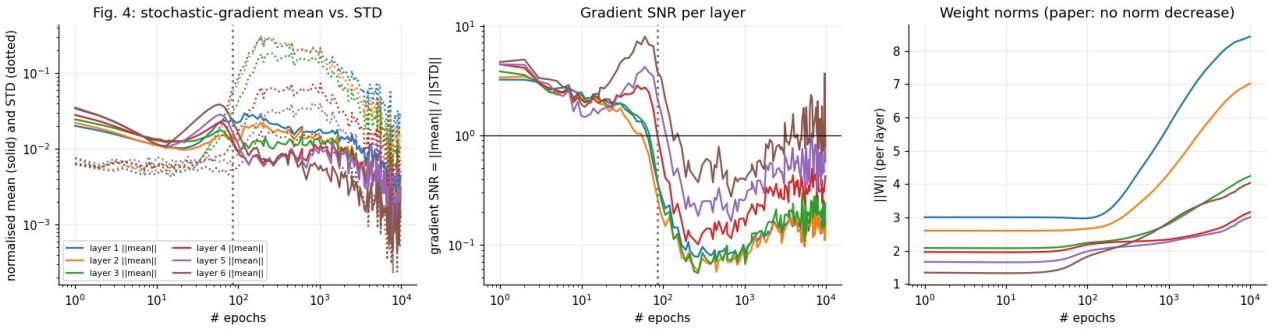


Figure 4 reproduction: gradient SNR, information, cross-entropy, and weight-norm diagnostics.

Diagnostic	Reproduced value	Interpretation
Initial geometric-mean SNR	3.97	The shared gradient direction initially dominates.
First post-peak SNR < 1	Epoch 84	Operational drift-to-diffusion transition.
Epochs after transition	About 99%	Most recorded training occurs in the low-SNR regime.
Paper transition	About epoch 350	Qualitative agreement, quantitative mismatch.
Weight norms	Increase	Compression is not explained by shrinking weights.

Using epoch 84 as the reference, layers 3 through 6 lose approximately 0.80, 2.83, 4.95, and 3.13 bits of binned input information by the end of training. Layer 1 remains at the ceiling and layer 2 changes by about 0.06 bit. Four of six layers meet the report's practical compression threshold. The SNR change and information change occur together, but the experiment does not identify a causal mechanism.

6 Information Bottleneck frontier

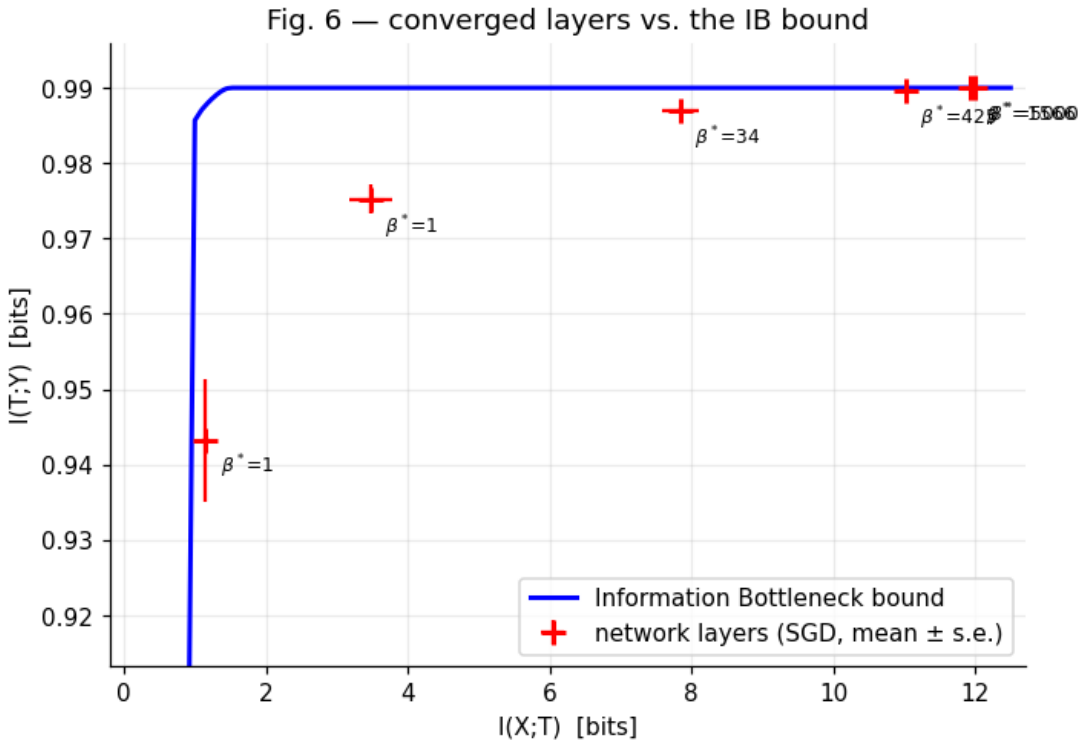


Figure 6 reproduction: converged layer locations and the computed Information Bottleneck frontier.

Quantity	Value	Scientific meaning
Mean vertical gap	0.010 bit	Learned layers retain nearly the frontier label information at their input coordinate.
Maximum vertical gap	0.044 bit	No endpoint has a large vertical prediction shortfall.
Layer 1 horizontal excess	10.55 bits	The widest layer remains highly redundant despite a small vertical gap.
Layers 1-3 horizontal excess	9.66-10.55 bits	Vertical proximity does not imply economical compression.
Output horizontal excess	0.20 bit	The output is close in both relevance and compression.

The frontier analysis gives the paper technical credit while refining its interpretation. Deeper layers approach an efficient trade-off. Wide layers can lie almost vertically on the curve and still carry many unnecessary input bits. The fitted beta values decrease with depth, but boundary values in the beta search grid should not be treated as precise physical constants.

7 Network depth

Fig. 5 — information paths for different numbers of hidden layers

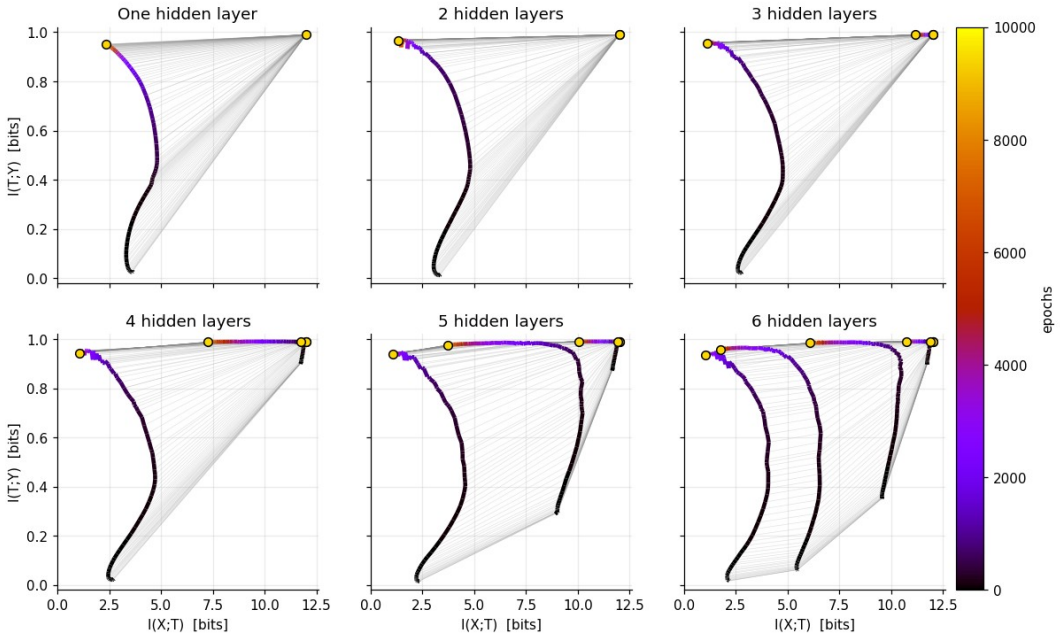
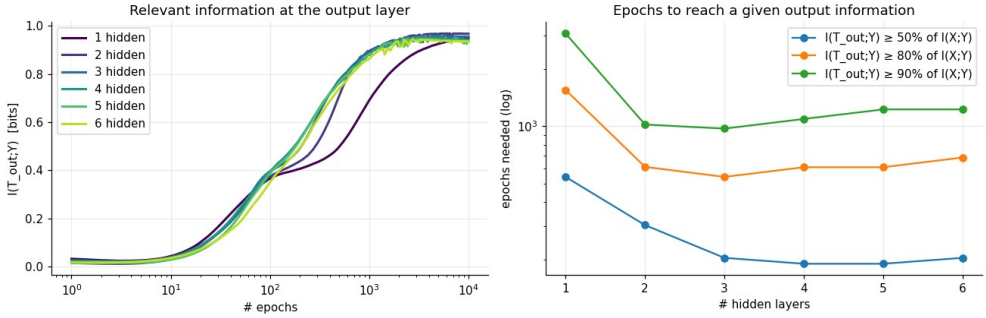


Figure 5 reproduction: threshold-crossing behavior for one through six hidden layers.



Epochs required to reach fixed fractions of available label information.

Hidden layers	Epochs to 80%	Relative conclusion
1	1,548	Slow baseline
2	612	Large improvement
3	542	Best observed time
4	609	No further gain
5	609	No further gain
6	685	Slower than three layers

The best reduction is approximately 2.9 times from one to three hidden layers. The sequence then becomes non-monotone. Epoch count also differs from wall-clock time and total floating-point work because deeper models perform more computation per epoch. The evidence supports a depth benefit over the shallow baseline, not a universal super-linear acceleration law.

8 Training-set size and task generality

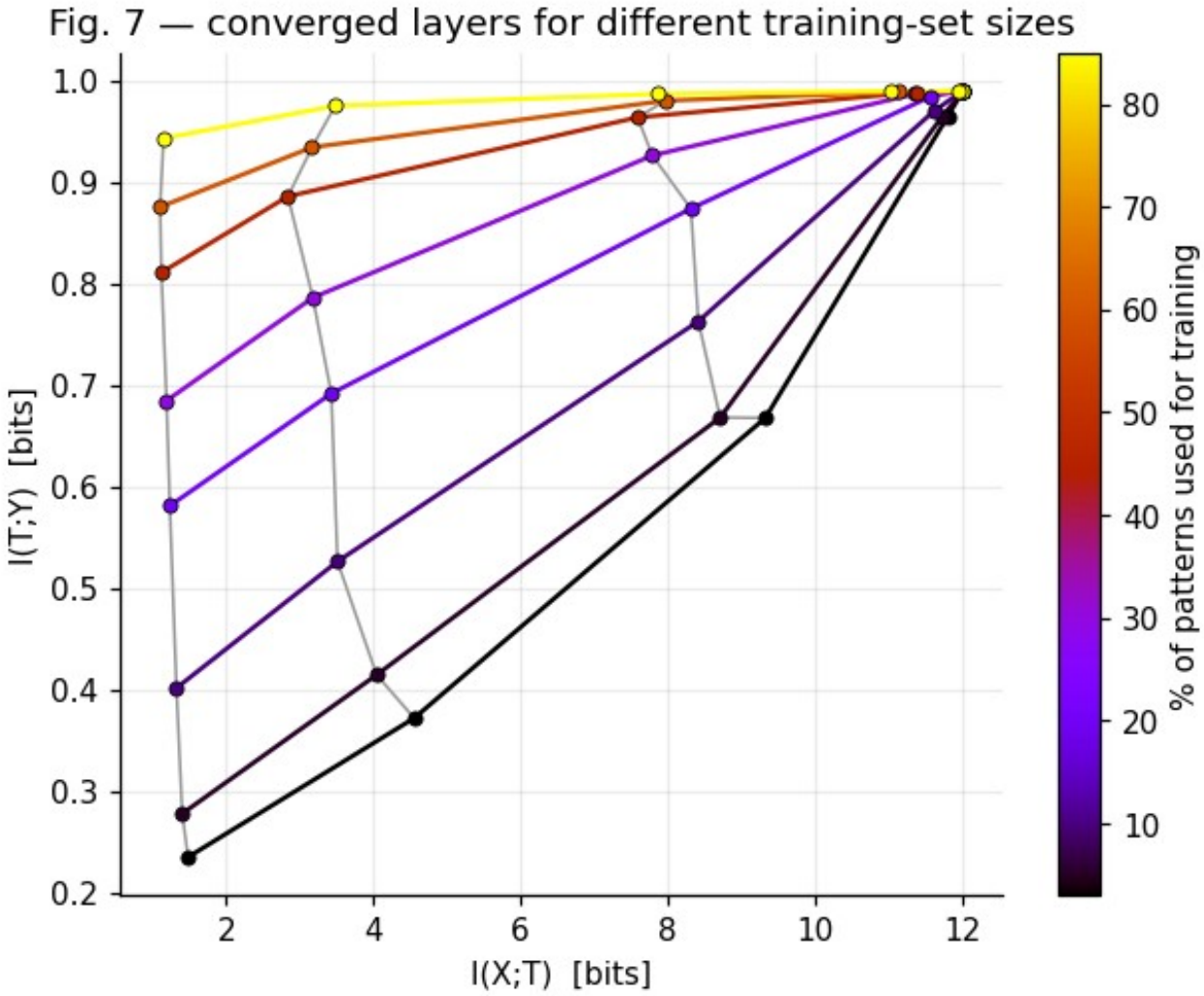


Figure 7 reproduction: final output-layer target information versus the fraction of the input universe used for training.

Training fraction	Final output I(T;Y) bits
3%	0.234
5%	0.277
10%	0.401
20%	0.581
30%	0.685

45%	0.811
60%	0.876
85%	0.943

The output approaches the task ceiling as the observed fraction increases. Because every value uses the full 4,096-pattern evaluation, the graph measures population-level retained label information rather than training-set accuracy.

Non-symmetric committee machine

Fig. 8 — non-symmetric committee-machine rule

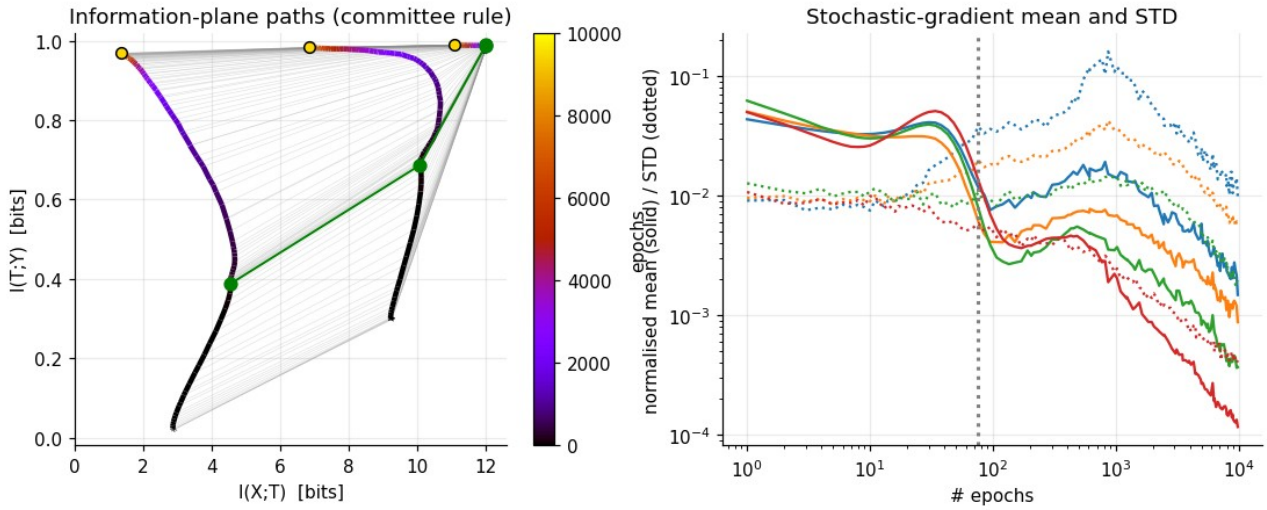
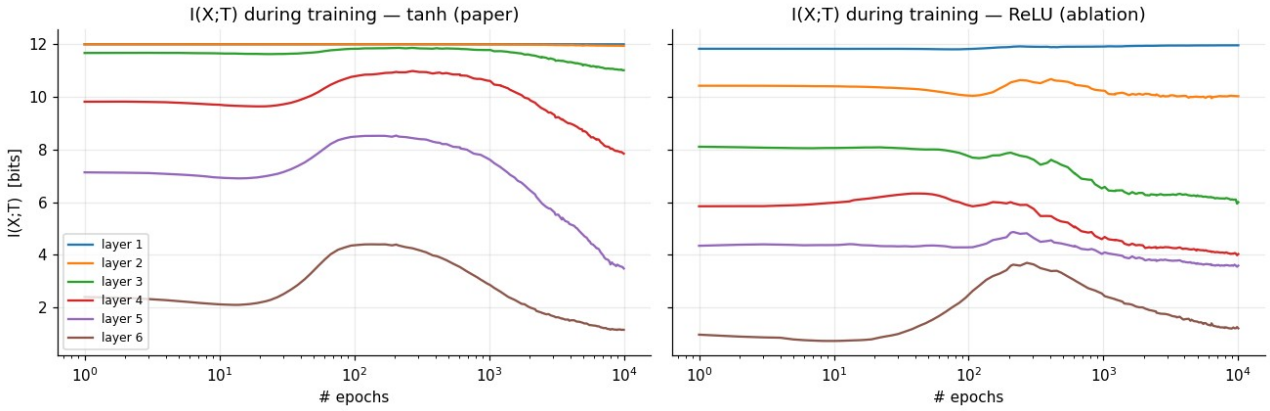


Figure 8 reproduction: committee-machine trajectories and gradient diagnostics.

The teacher is a majority vote of five random perceptrons, and the student network is 12-8-5-3-2. The stochastic rule is calibrated to the same 0.990-bit information ceiling. The SNR transition occurs near epoch 75 and the layers again rise before moving left. Similarity across two synthetic tasks supports qualitative robustness, while the limited task family prevents a broad claim about arbitrary architectures or real-world data.

9 Additional experiments

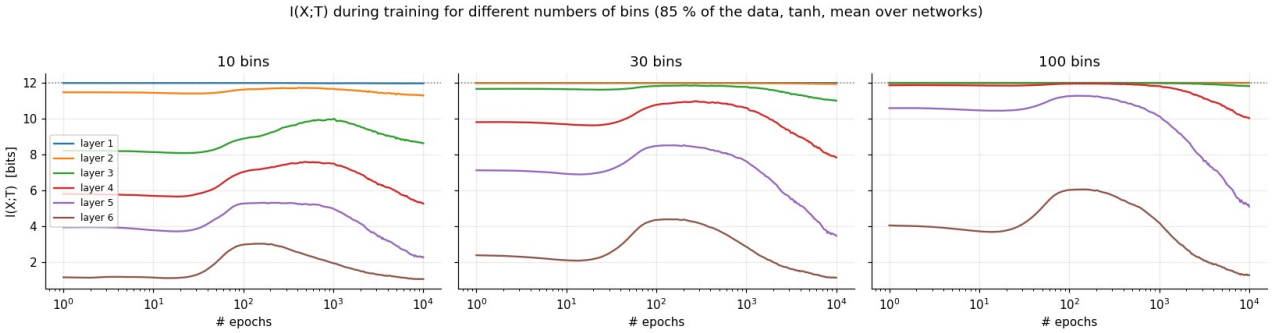
Tanh and ReLU



Peak-to-final binned information changes for tanh and ReLU networks.

Four of six tanh layers and five of six ReLU layers satisfy the practical drop criterion in the submitted analysis. This does not show that ReLU networks undergo the same intrinsic compression. ReLU activations are unbounded, and the implementation rescales bins to the current [0,max] range at each snapshot. A growing maximum can merge ordinary observations into fewer bins even when their continuous geometry remains informative. A fixed-range control is required.

Bin-count sensitivity



Information estimates at 10, 30, and 100 bins.

The number of layers meeting the compression criterion changes from five at 10 bins to four at 30 bins and three at 100 bins. Layers 4 through 6 lose information under all three settings, but the magnitude changes substantially. At 100 bins, layers 1 and 2 reach the 12-bit ceiling. The direction is robust for selected deeper layers; the estimator is not invariant.

10 Comparison with the original paper

Claim	Original paper	Submitted reproduction	Assessment
Fitting and compression	Two-stage information-plane motion	Clear in deeper layers under 30-bin quantization	Qualitative agreement
Late diffusion	Transition around epoch 350; late phase dominates	Operational SNR crossing near epoch 84; about 99% late	Pattern agrees; timing differs
IB frontier	Layers approach the bound	Vertical gaps small; wide layers retain large horizontal excess	Qualified agreement
Depth	Additional layers accelerate training strongly	Improvement to three layers, then non-monotone	Partial reproduction
Generality	Additional task shows similar motion	Committee task crosses near epoch 75	Limited robustness evidence

11 Limitations and threats to validity

- Quantization bias: the reported mutual information combines network dynamics with the selected binning rule.
- Continuous representations: a lower binned estimate does not prove an equal decrease in continuous mutual information.
- Operational diffusion: an SNR crossing does not establish Brownian dynamics or causal diffusion.
- IB optimality: a small vertical gap does not prove global optimal compression, especially when horizontal excess is large.
- Ensemble size: the reproduction uses 20 networks rather than the paper's 50.
- External validity: the main evidence comes from small synthetic tasks and manual NumPy networks.
- Critical slowing down: the notebook does not test the paper's critical-slowing-down claim.

12 Conclusions

The project reproduces the main information-plane trajectory and connects it to a transparent implementation of data generation, NumPy backpropagation, SGD, quantized mutual-information estimation, and gradient diagnostics. The evidence is strongest for fitting followed by lower binned input information in deeper layers and for the effect of training coverage on retained target information. The IB and depth claims require two-dimensional and non-monotone interpretations. The ReLU and bin-count studies demonstrate that measurement choices are part of the scientific result.

A stronger follow-up should fix quantizer ranges across epochs, report exact activation collisions on the finite input universe, compare multiple estimators, repeat across independent seeds and SGD hyperparameters, match models by computation as well as depth, and evaluate larger tasks. These controls would separate network behavior from estimator behavior more sharply.

Reproducibility resources

The interactive website, attribution-corrected notebook copy, silent demonstration video, timestamped presenter script, PowerPoint with embedded notes, and validation manifest form the reproducibility package. The verified public explorer is available at https://arba318.github.io/itct-midterm/. Its published source is stored in the public GitHub Pages repository at https://github.com/arba318/arba318.github.io/tree/main/itct-midterm. The first verified deployment of this final research site was produced from repository commit 1f89125. The validation manifest records the live-site checks, local tests, artifact checksums, and the distinction between notebook-supported evidence and qualified interpretation.

References

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